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**UNIVERSITY
OF THE YEAR**

Model-Based Testing

Software Testing (COM3529)

Checkin code: JD-YG-GH

José Rojas & Neil Walkinshaw

Week 7

Roadmap

Beizer's Maturity Model

Test Automation

Unit Testing

Control / Data-Flow Analysis

Code Coverage

Regression Testing

Model-Based Testing

Mutation Testing

Search-Based Test Generation

Fuzzing

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Beizer's Maturity Model

Regression Testing

Test Automation

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Unit Testing

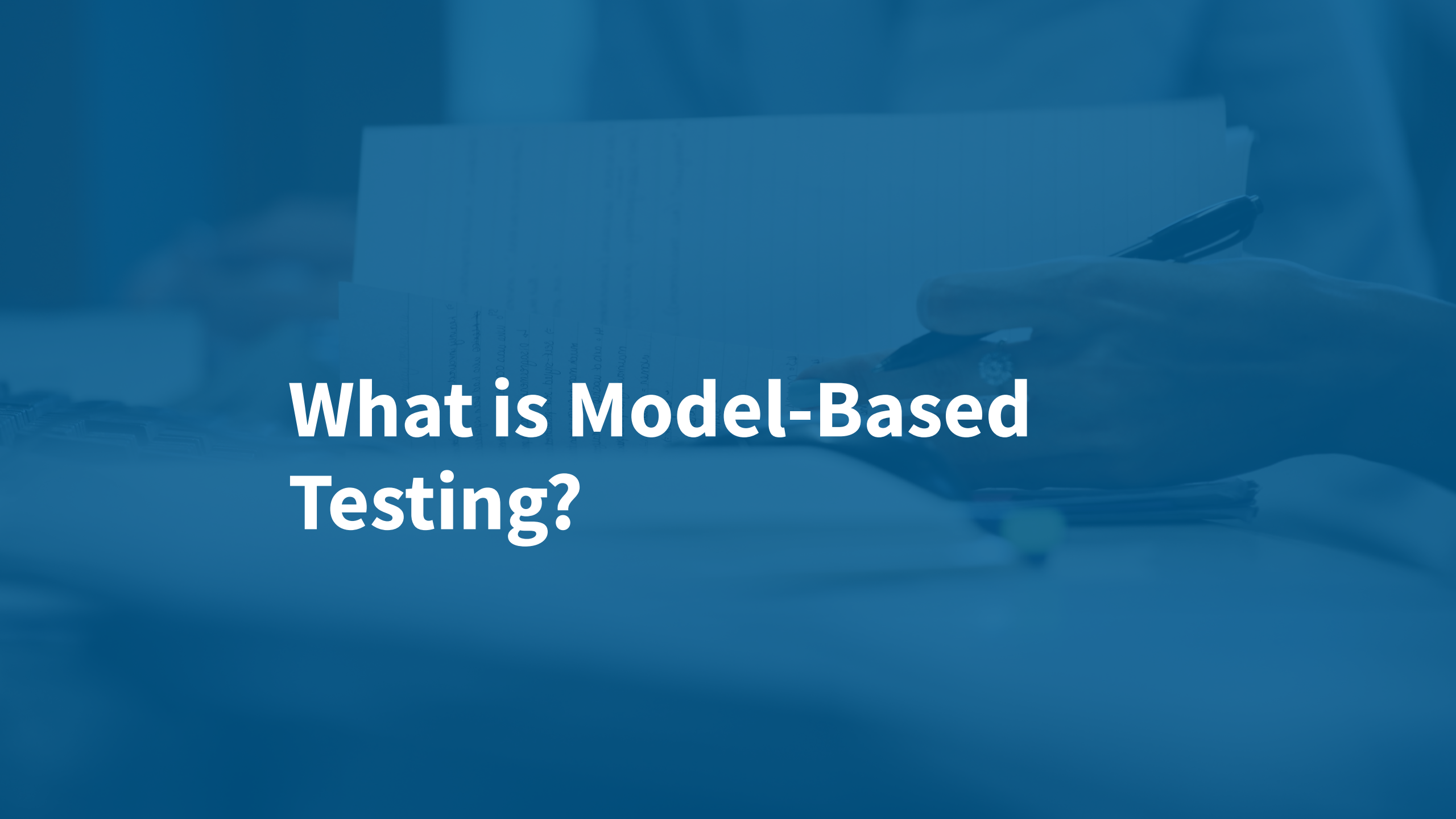
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A hand holding a pen over a document with a blue overlay.

What is Model-Based Testing?

Models of Software Behaviour

Formally capture *intended system behaviour*.

Abstractions of the low-level source code that capture key behaviours.

Sometimes even the main development artefact (Model-Based Development)

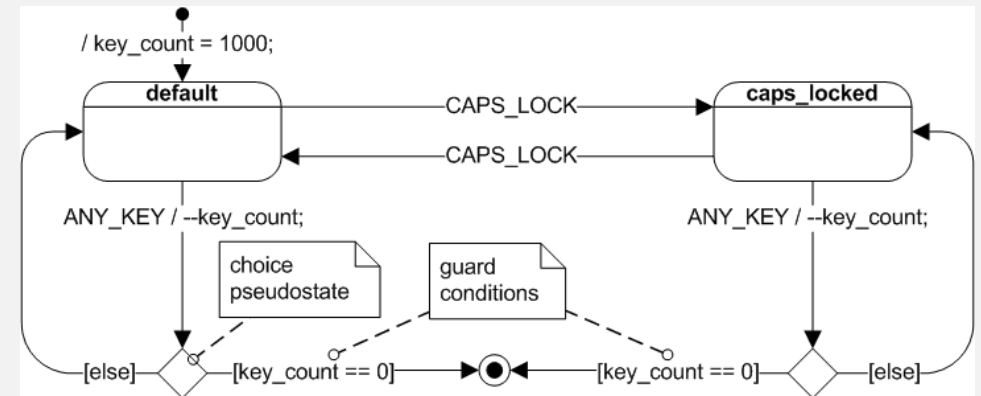
Many different types of models. E.g.:

State-based models - constraints on the data-state before / after a function has executed.

Transition-based models - potential sequencing of inputs (sometimes corresponding outputs).

```
procedure Increment (X : in out Counter_Type)
  with Global => null,
       Depends => (X => X),
       Pre      => X < Counter_Type'Last,
       Post     => X = X'Old + 1;
```

Example of pre-/ post-conditions in SPARKAda



Example of a UML state machine

Models are extremely valuable for *testing*

Consider the key testing challenges:

Find inputs that explore key behaviours.

Determine whether or not outputs are correct.

Decide when we have tested enough.

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Decide when we have tested enough.

Models offer basis for assessing coverage.

The background of the slide features a blurred image of a person's hand holding a pen and writing on a notepad. The entire image is covered with a semi-transparent blue filter. The title text is centered over this background.

Finite State Machines (FSMs)

Definition of a Finite State Machine

An FSM is a sextuple $(S, s_0, X, Y, \delta, \lambda)$ where:

S is a finite set of states

$s_0 \in S$ is the initial state

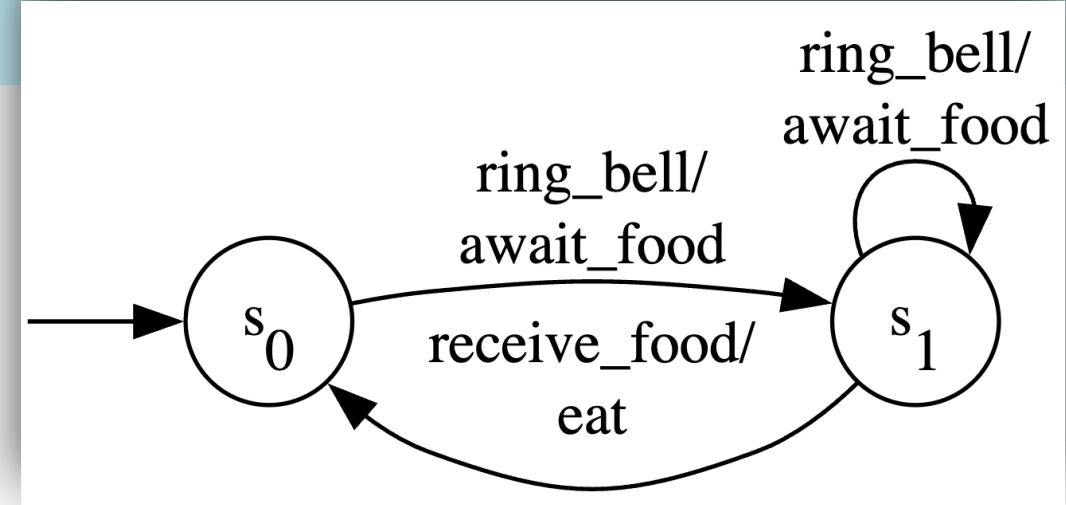
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Can extend δ and λ to take sequences, giving δ^* and λ^*



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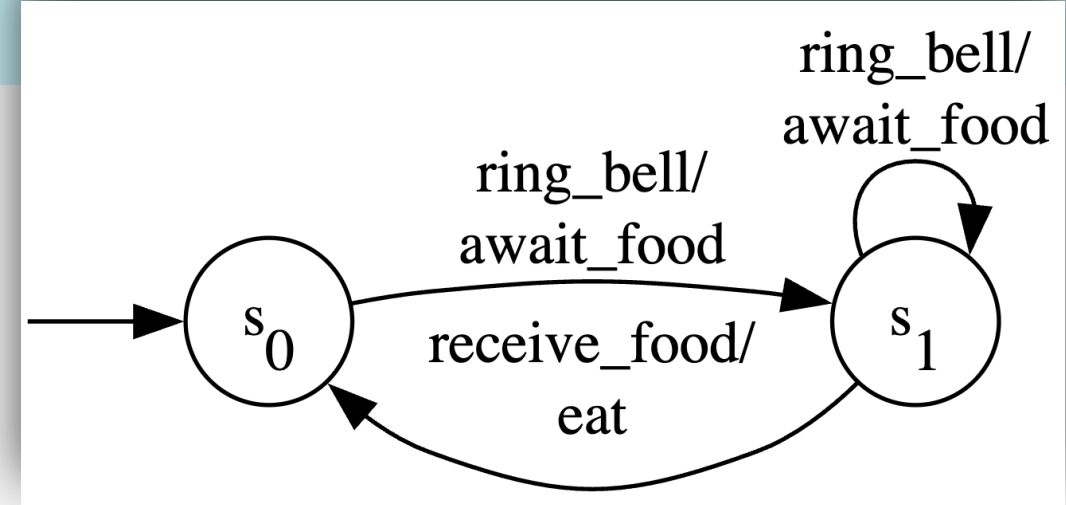
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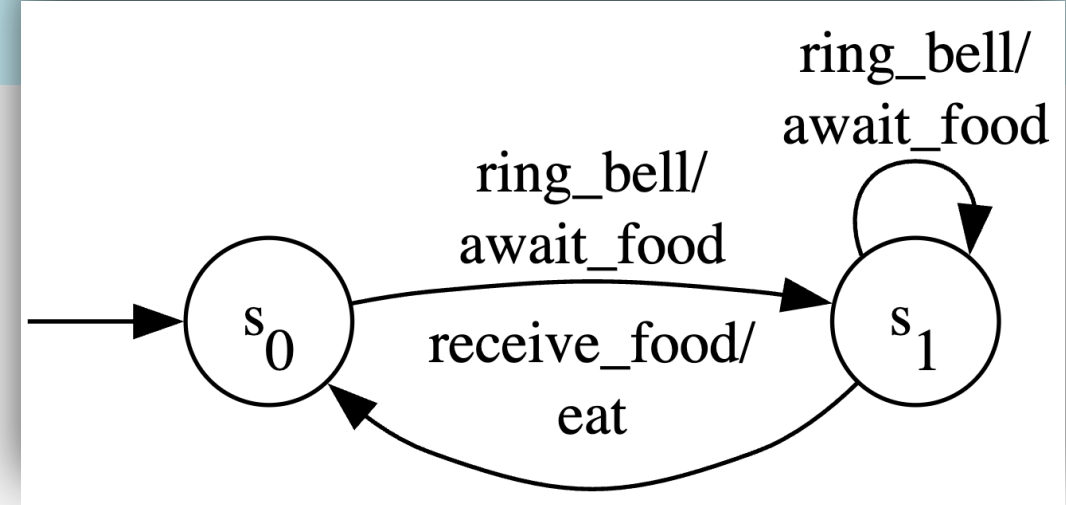
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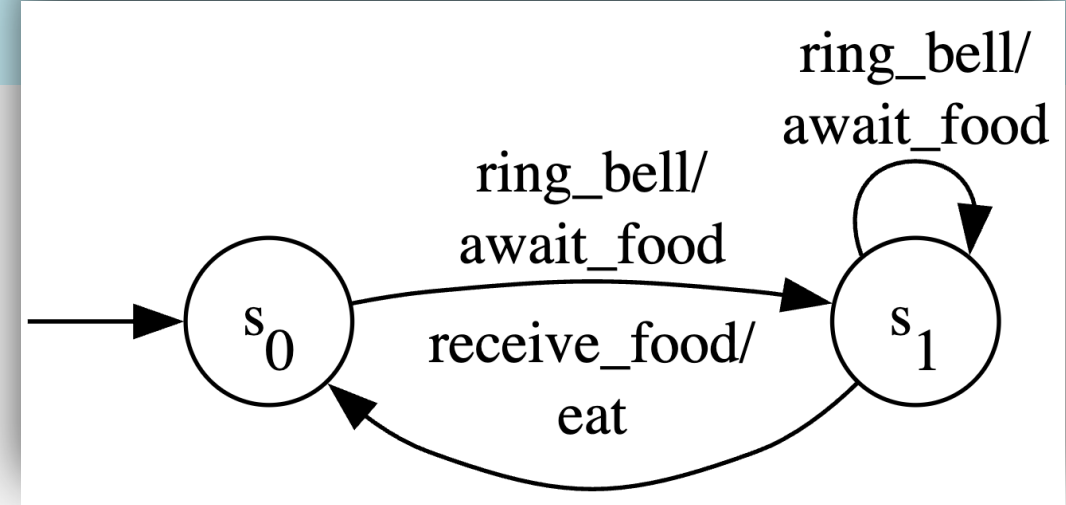
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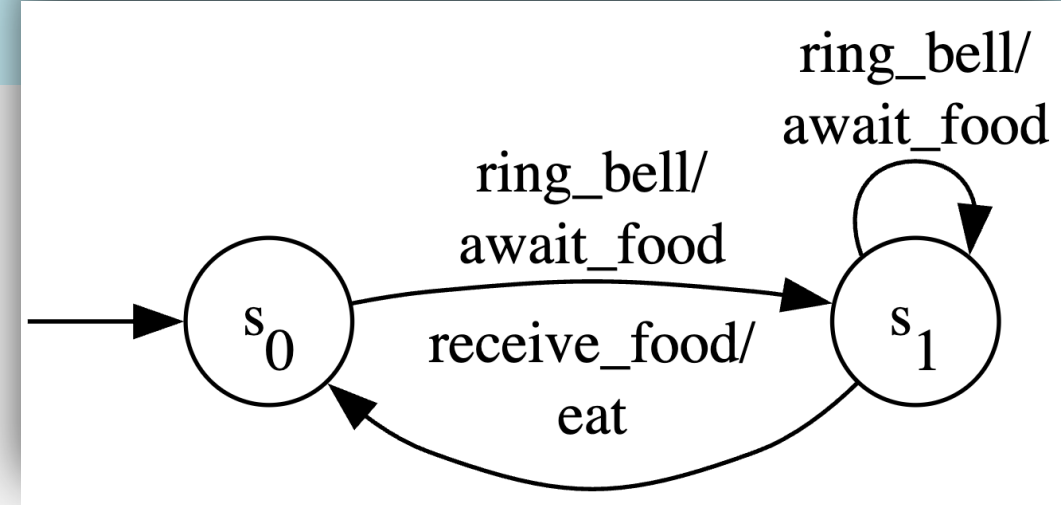
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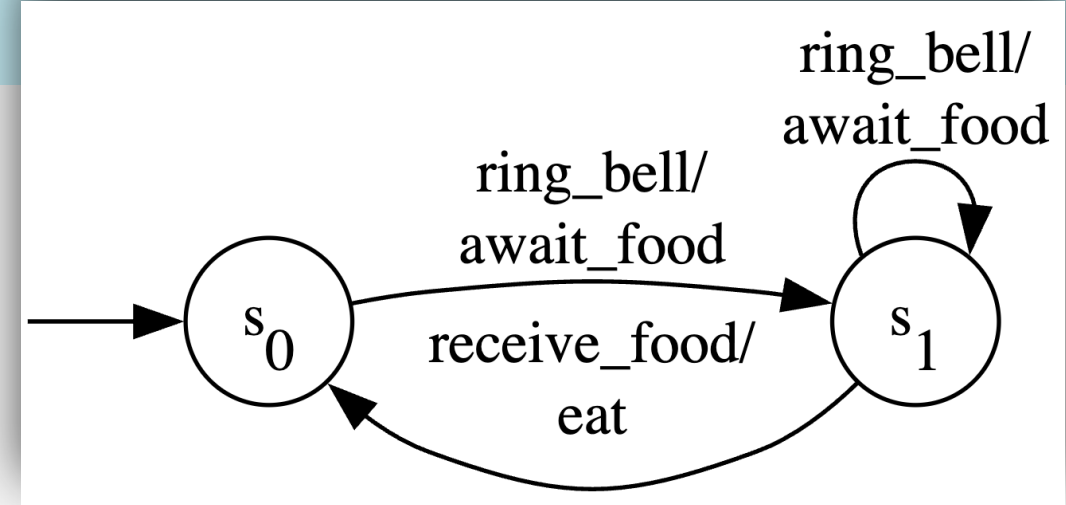
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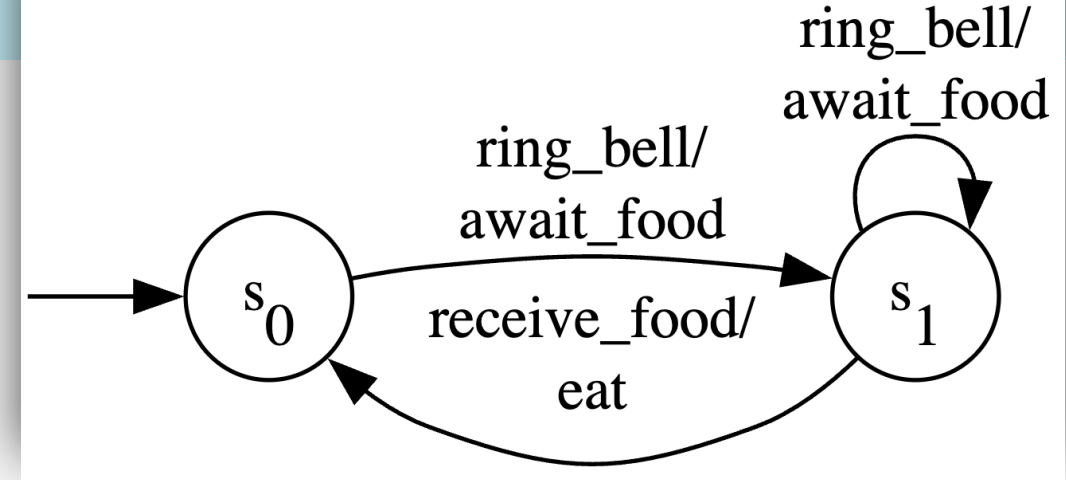
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Properties of FSMs

An FSM is **deterministic** if there is at most one transition from each state for each input.

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I.e. No smaller FSM in terms of number of states defines the same regular language.

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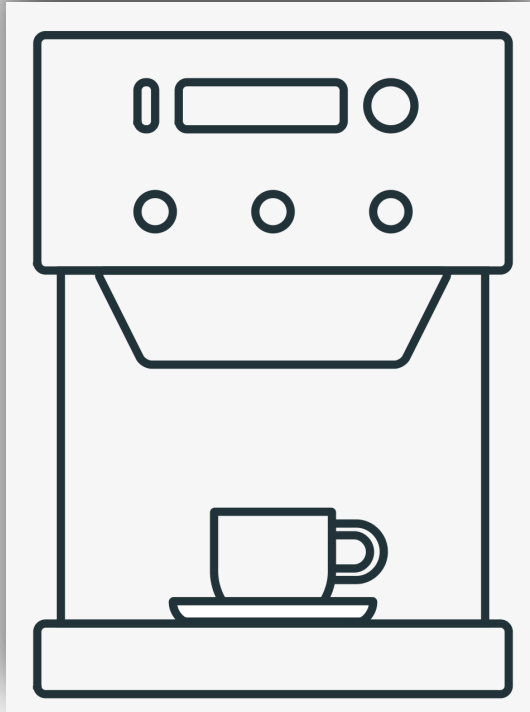
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An FSM is **strongly connected** if, for each pair of states (s_i, s_j) there exists an input sequence that leads from s_i to s_j .

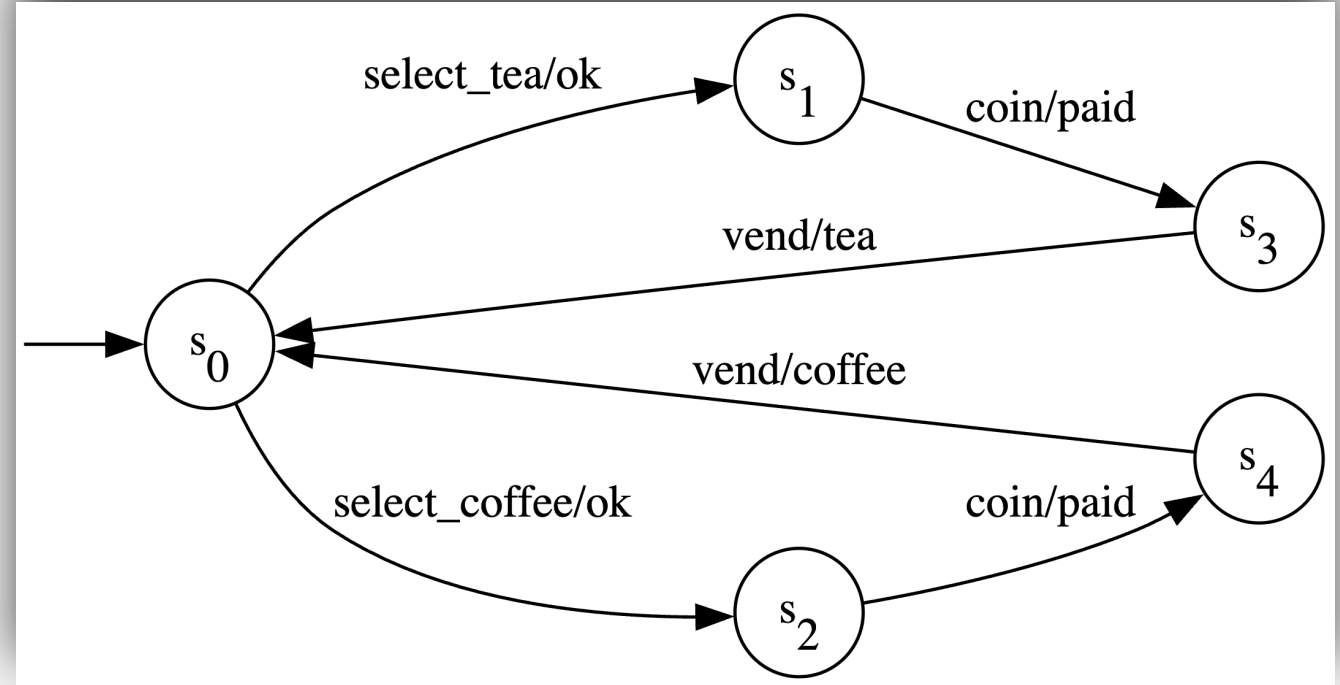
A hand holding a pen is positioned over a document, with a blue overlay covering the entire image. The text "Testing with FSMs" is centered in white.

Testing with FSMs

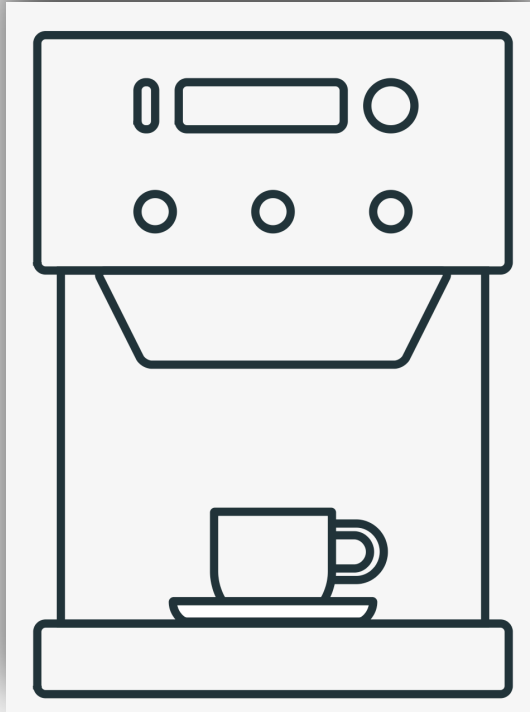
FSM Testing of a Coffee (or Tea) Vending Machine



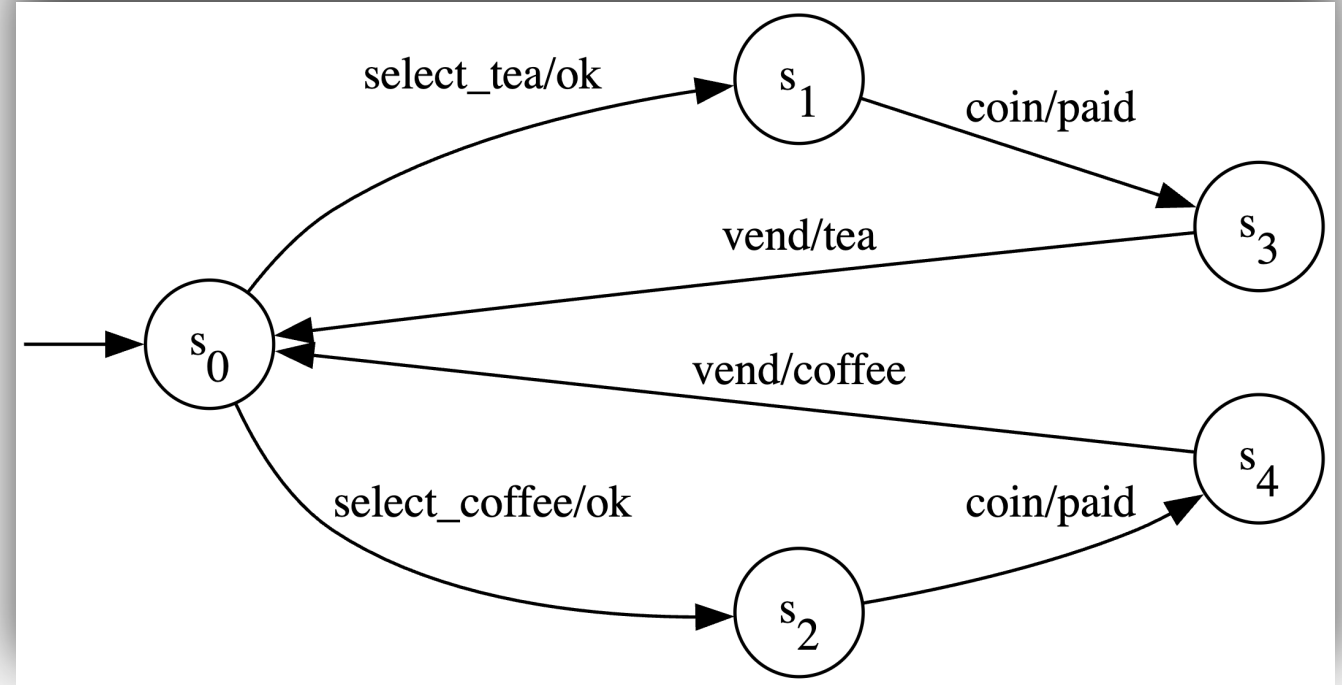
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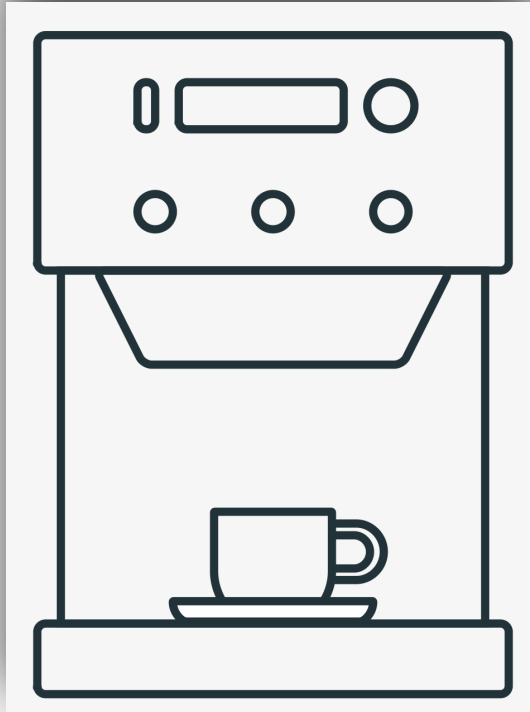


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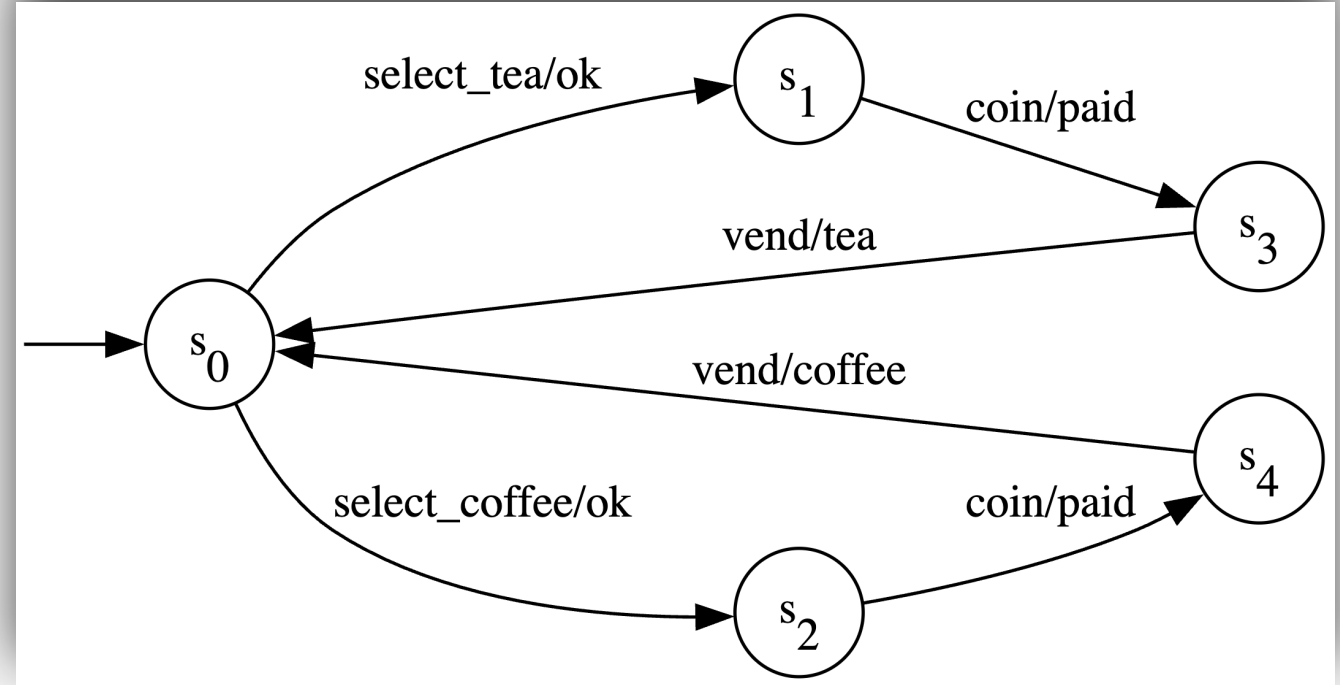


Assume that the system behaves like an FSM model.

FSM Testing of a Coffee (or Tea) Vending Machine



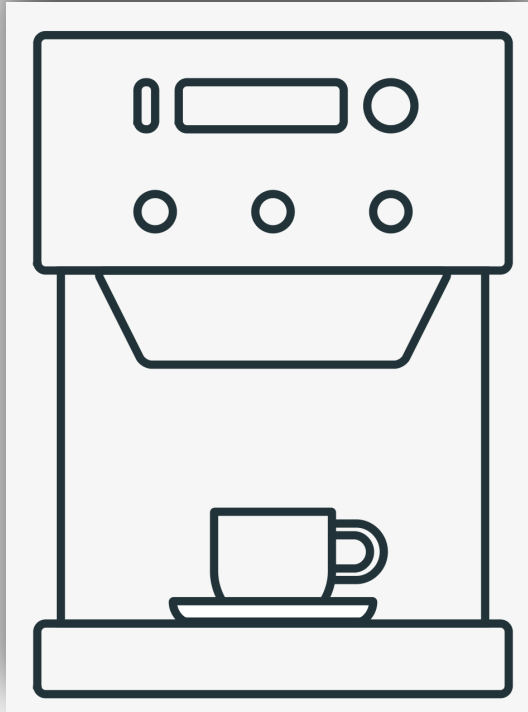
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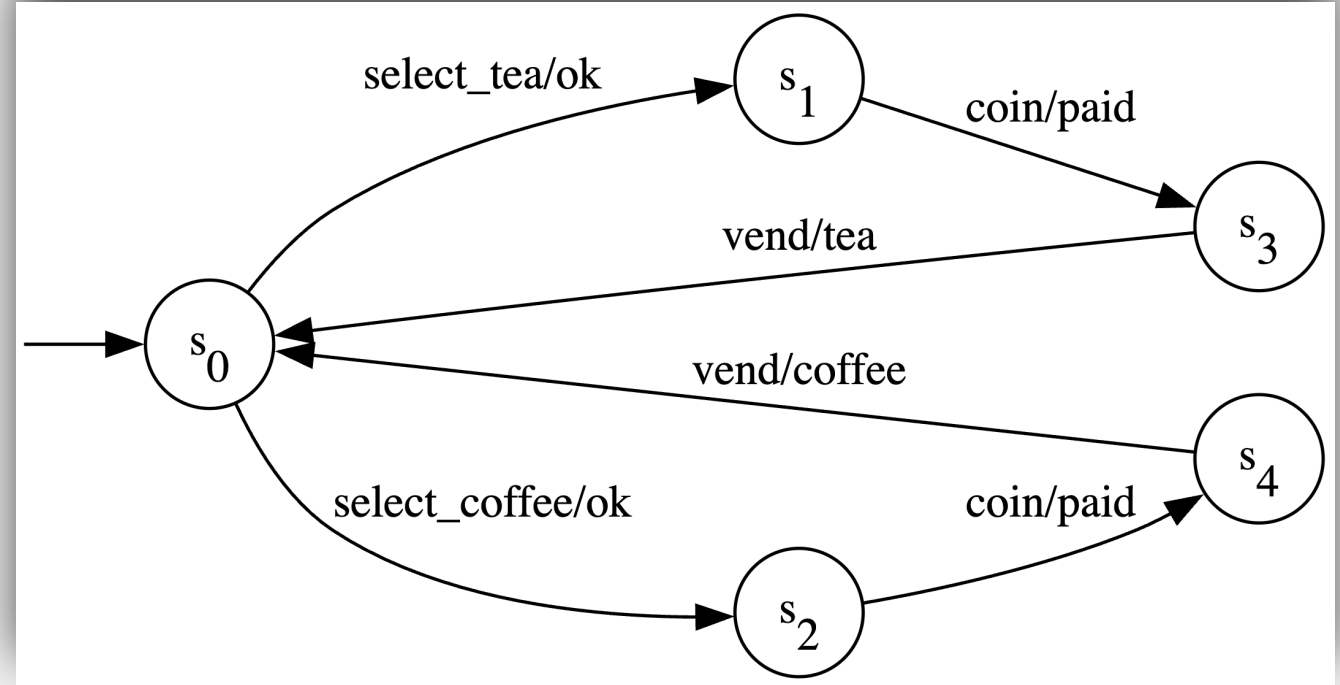
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Use our model to derive a test set (a set of input sequences).

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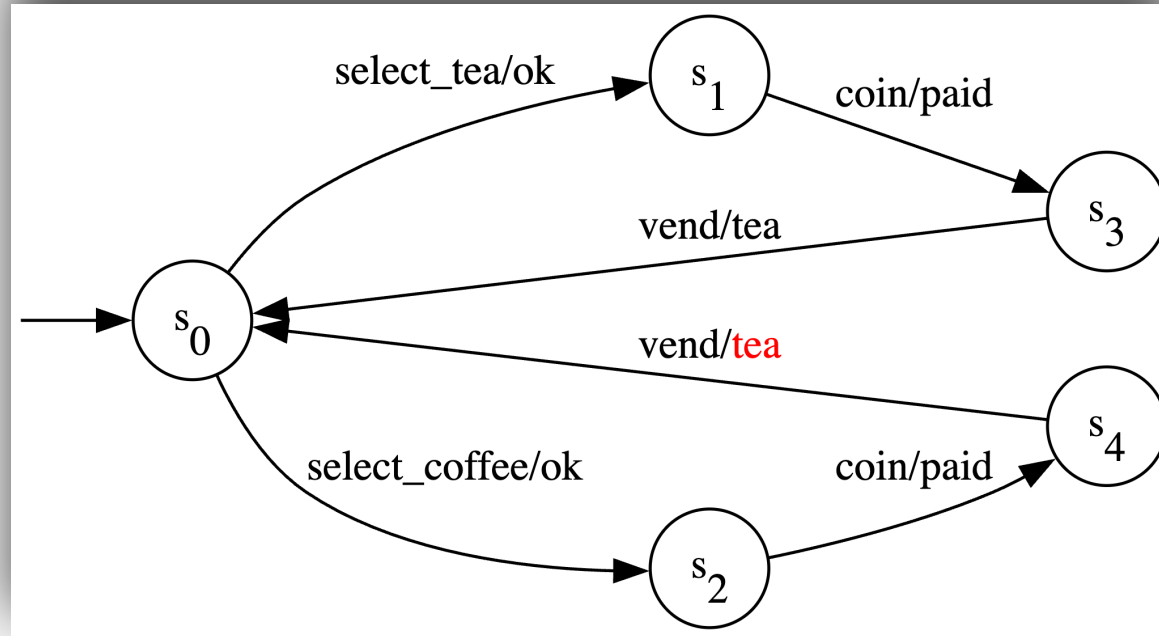
Use our model to derive a test set (a set of input sequences).

Observe corresponding behaviour in the system, and compare outputs.

Output faults

An **output fault** occurs when a transition produces the wrong output.

Often the easiest to find - just need to execute the faulty transition to observe an erroneous output.

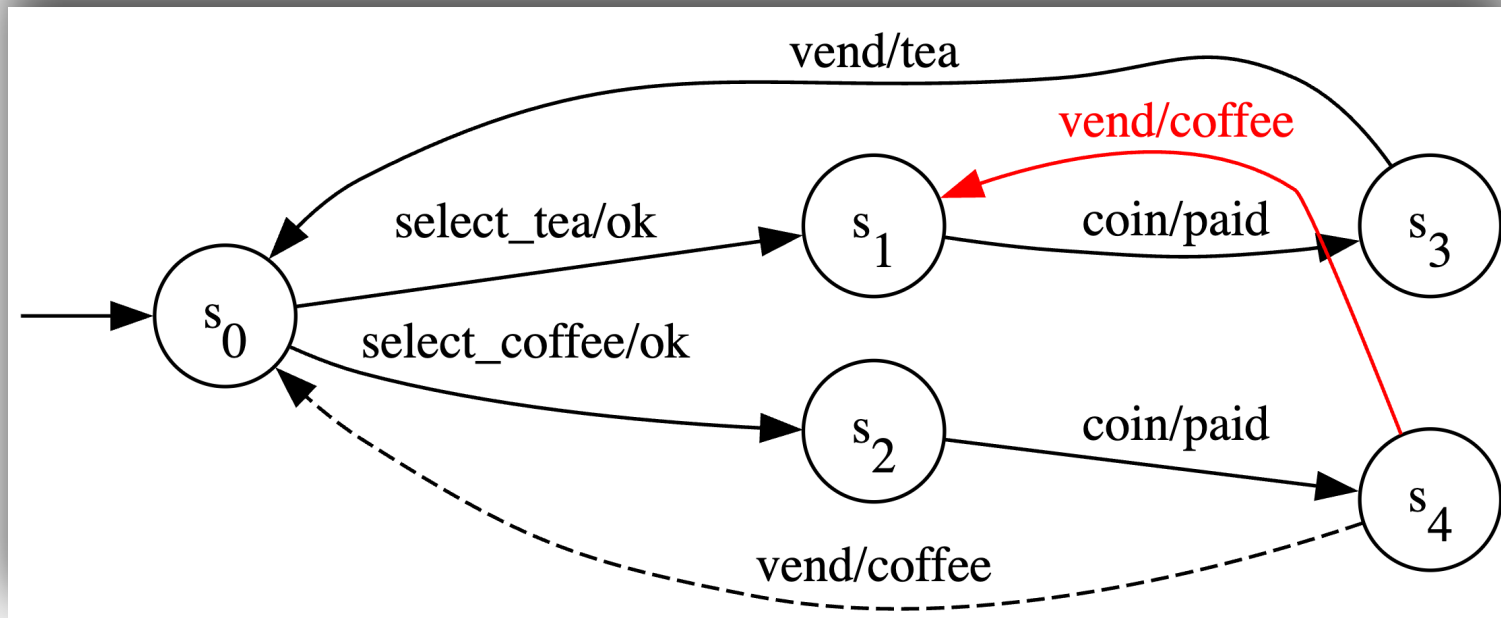


State transfer faults

A **transition fault** occurs when transition leads to the wrong state.

Often the easiest to find - just need to execute the faulty transition to observe an erroneous output.

Can also indicate **extra states** in the implementation that do not exist in the model





Detecting output faults with Transition Tours

The Transition Tour Method

Find a path from s_0 that includes **all transitions** in the model.

If the FSM is strongly connected, this can always be achieved in a single path.

Otherwise it might require a set of paths.

A **test sequence** is the sequence of inputs and outputs that corresponds to a path.

During a test, we ensure (1) that the input is possible, and (2) that the output from the system matches the output in the model.

Guaranteed to find all output faults, as long as there are no state transfer faults.

Plenty of graph-theoretical algorithms that can efficiently identify a tour.

Finding the transition tour is equivalent to the “Chinese Postman Problem” in Graph Theory.

Heuristic approach to obtain Transition Tour

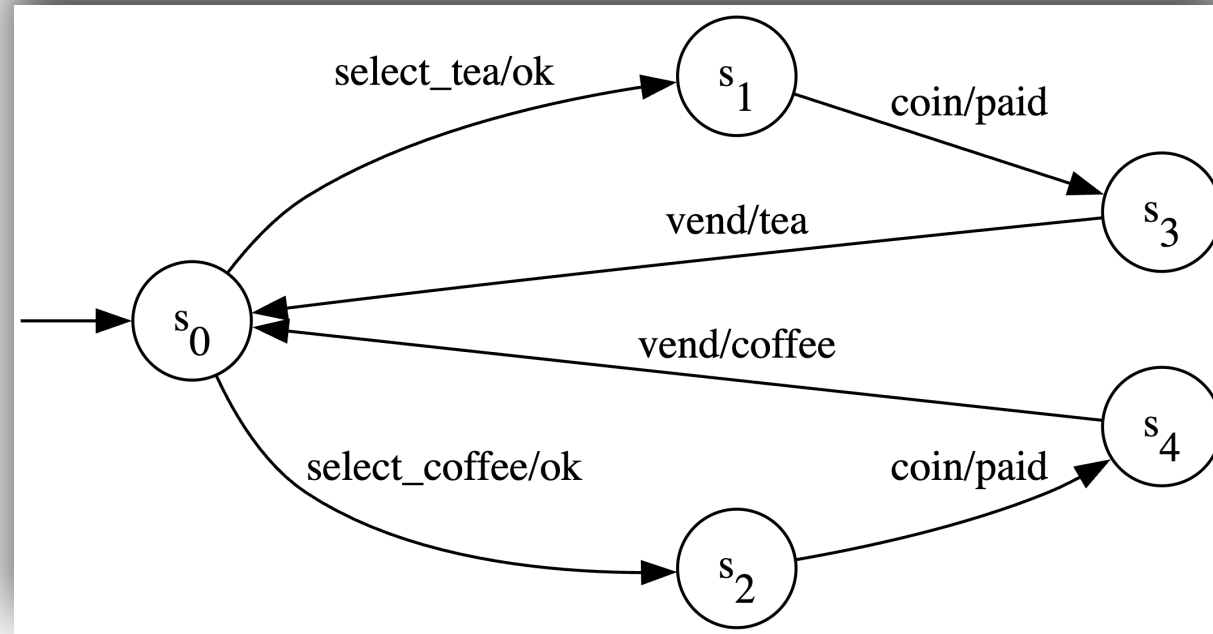
Heuristic approach:

In current state s , if there is a transition t that has not been included:

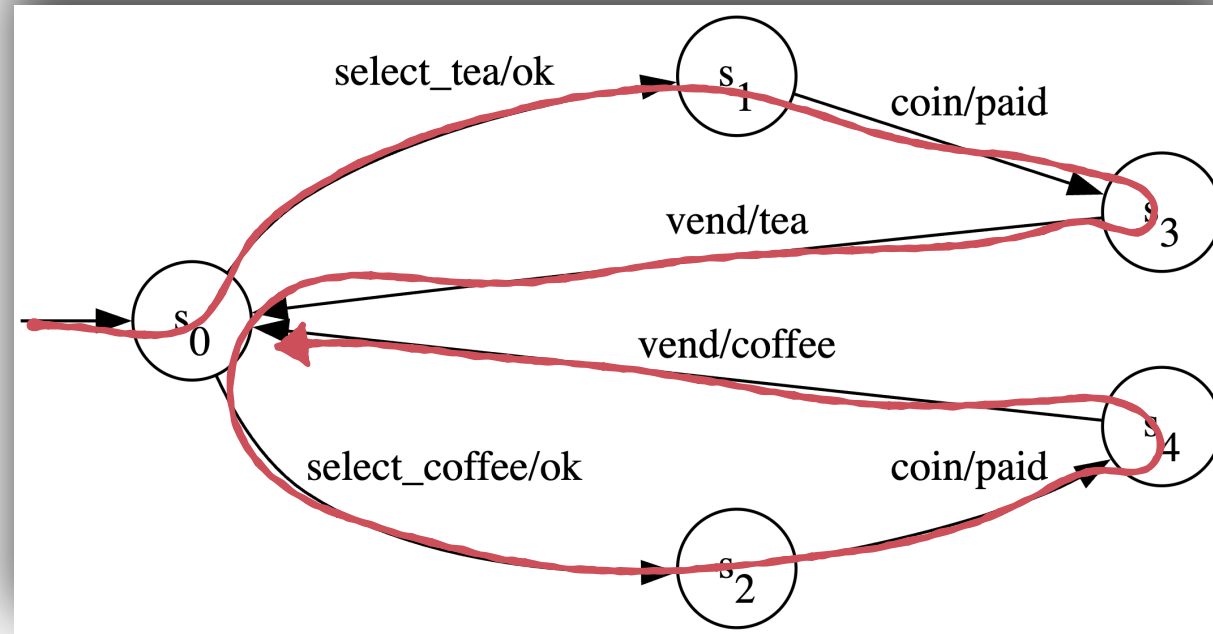
add t to current path and update state.

Otherwise follow shortest path from s that reaches a state with at least one uncovered transition.

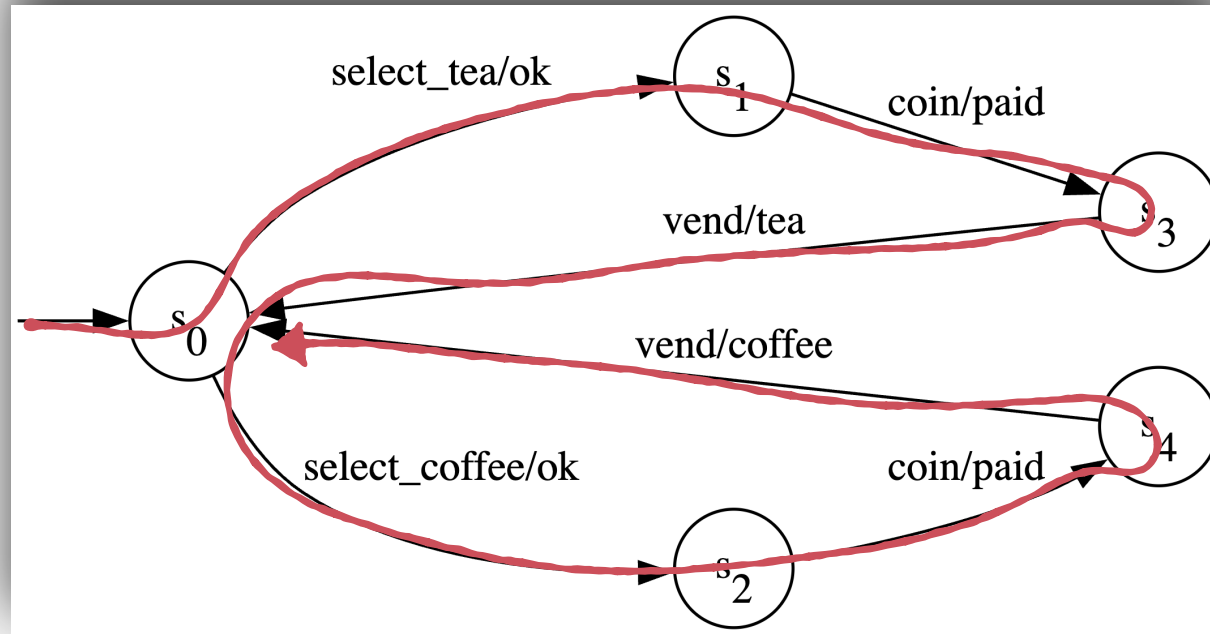
Testing the Drinks Machine



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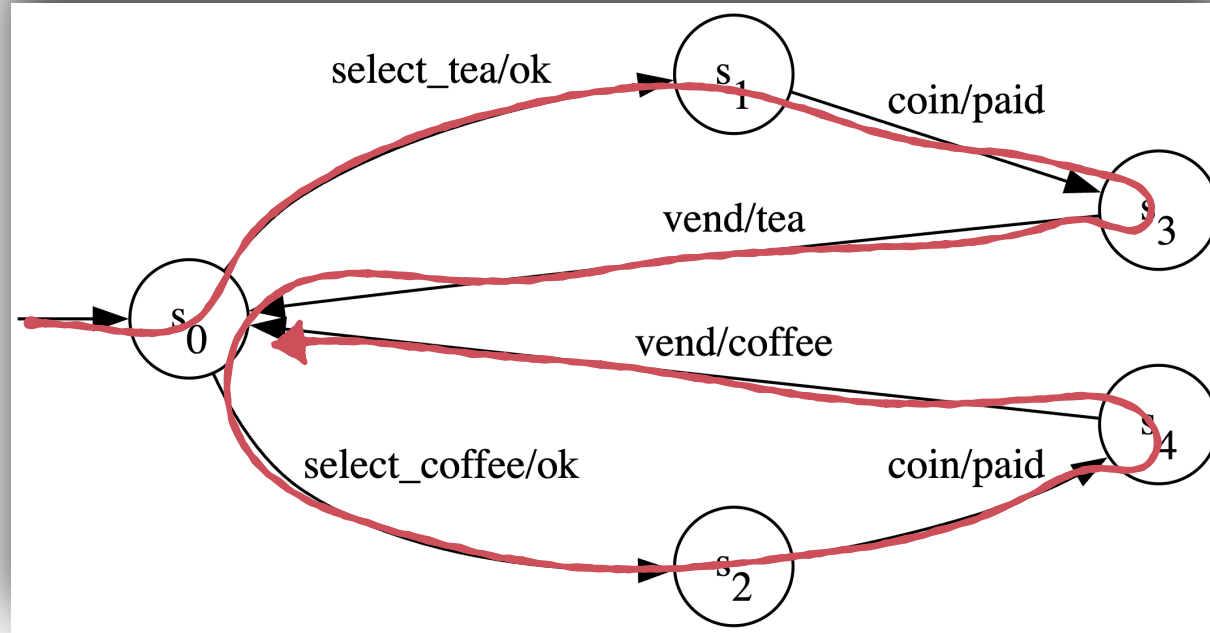
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Test sequence:

select_tea/ok, coin/paid, vend/tea, select_coffee/ok, coin/paid, vend/coffee

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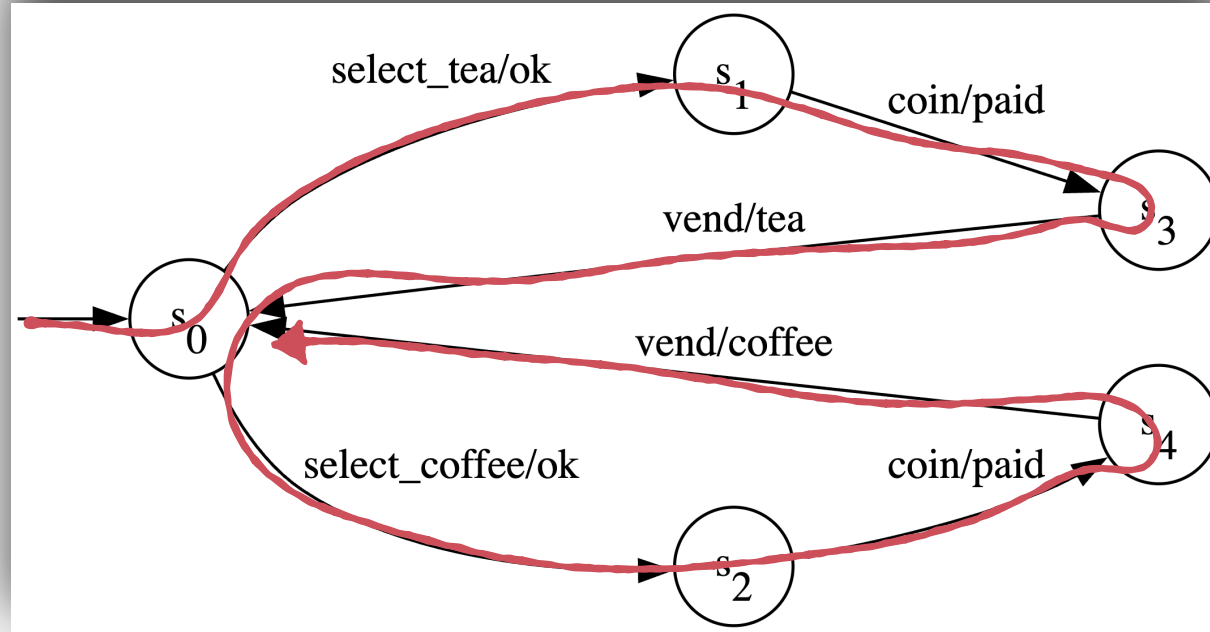


Test sequence:

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Input sequence: *select_tea, coin, vend, select_coffee, coin, vend*

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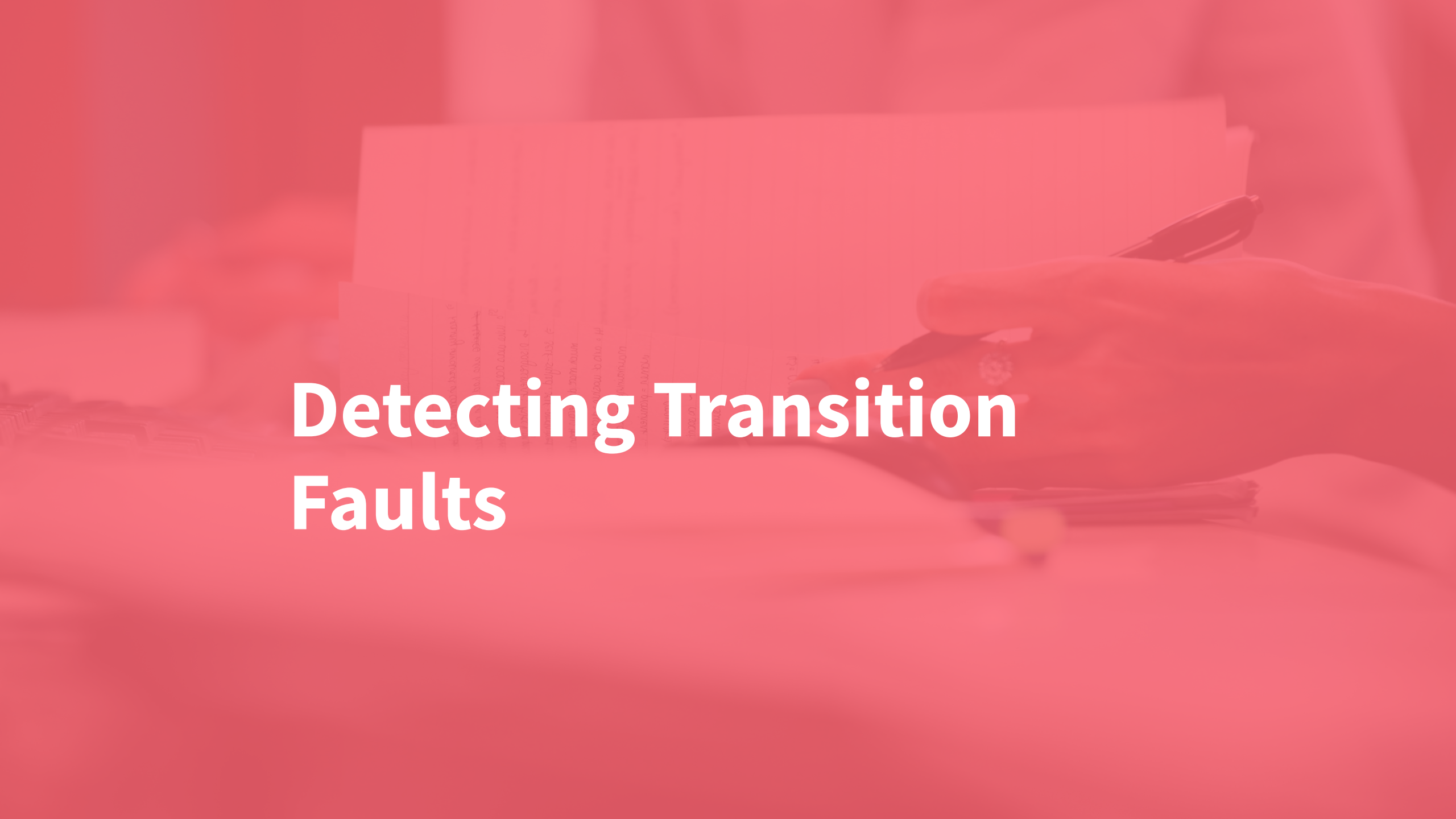


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select_tea/ok, coin/paid, vend/tea, select_coffee/ok, coin/paid, vend/coffee

Input sequence: *select_tea, coin, vend, select_coffee, coin, vend*

Output sequence: *ok, paid, tea, ok, paid, coffee*

A hand holding a pen is positioned over a document, with a red overlay covering the entire image. The text "Detecting Transition Faults" is written in white on the red background.

Detecting Transition Faults

Testing Challenge

To test for transition faults, we want to check every transition $(q_i, q_j, x/y)$.

Basic process:

- Get the system into state q_i

- Submit input x

- Check that we get output y

- Check that we have ended up in state q_j**

Essential challenge:

- Given that we can only supply inputs and observe outputs ...

- ... how can we be sure that the system is actually in state q_j ?

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Distinguishing sequences
Unique I/O sequences (UIOs)
Characterising set

What makes a pair of states different?

The same sequence of inputs returns different outputs.

Formally, two states s and s' are separated by an input sequence x if:

$$(\lambda^*(s, x) \neq \lambda^*(s', x))$$

If there is such a sequence x , then s and s' are said to be **separable**.

If two states are not separable, they are said to be **equivalent**.

Distinguishing Sequences

A distinguishing sequence D is an input sequence, producing different outputs **for each state**.

For every pair of states s and s' such that $s \neq s'$, we have that $(\lambda^*(s, D) \neq \lambda^*(s', D))$

This means that the output produced for D **identifies** the state of the FSM.

A FSM can have multiple distinguishing sequences, but does not necessarily have one.

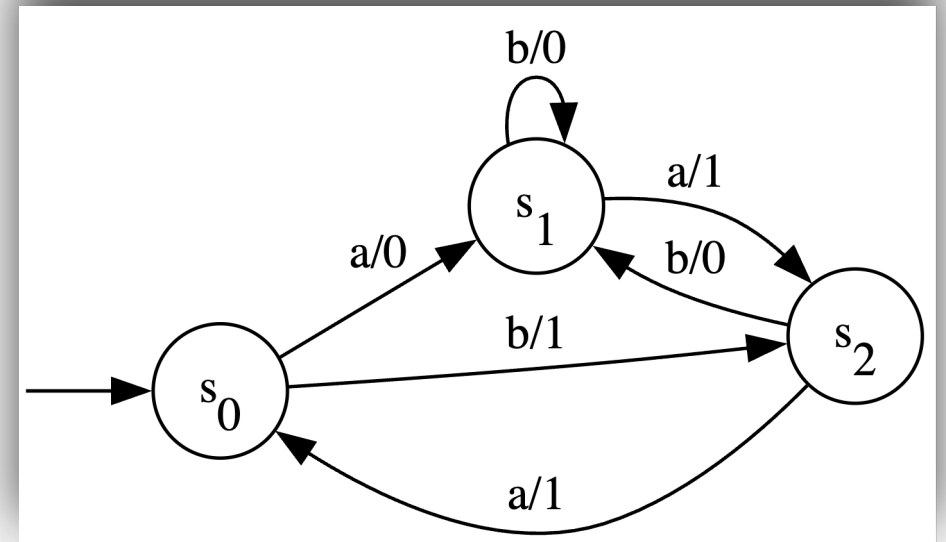
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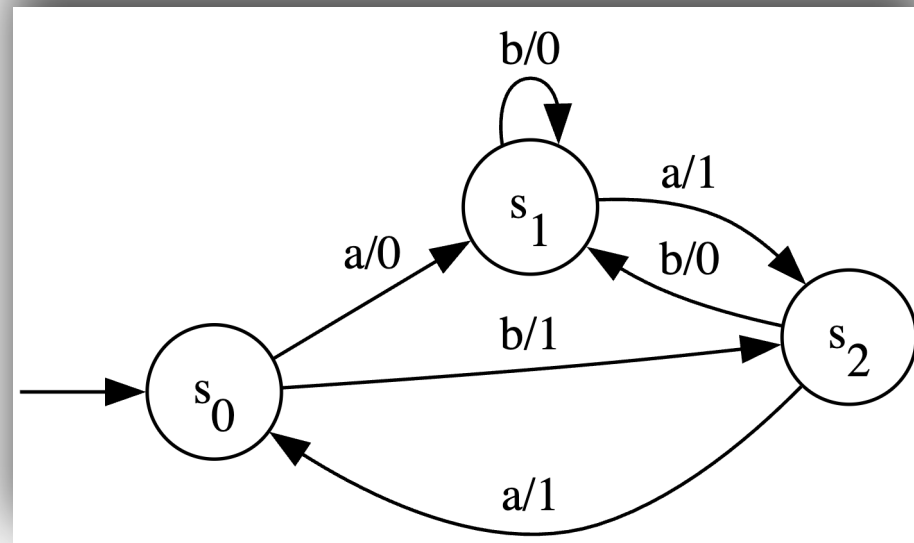
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For the example machine, we have $D = \langle a, a \rangle$

From s_0 the output sequence is $\langle 0, 1 \rangle$

From s_1 the output sequence is $\langle 1, 1 \rangle$

From s_2 the output sequence is $\langle 1, 0 \rangle$



Unique I/O Sequences (UIOs)

A UIO sequence separates **one state** from **all other states**.

To see if the FSM is in state s we can execute the UIO sequence for s .

Formally, $\lambda^*(s, \bar{x}) = \bar{y} \implies (\forall s'. s' \neq s \implies \lambda^*(s', \bar{x}) \neq \bar{y})$

$\bar{x}$ and $\bar{y}$ represent a sequence of inputs and outputs respectively.

If $\bar{x}$ produces $\bar{y}$ then we *must* have been in state s , otherwise we *must* have been in a different state.

Whereas a distinguishing sequence can identify *all* states, a UIO can only identify a single state.

Not all FSMs have UIOs either.

Brute-Force Approach

Brute-Force Approach

Systematically try out all possible input sequences of increasing length until UIO sequence is found for each state.

Steps:

- Start with input sequences of length 1, e.g. 'a', 'b'

- Apply each sequence to the FSM, starting in each state

 - Building a state / transition table can be helpful for this.

- Observe output sequences for each state.

 - If unique for a given state with respect *all* other states, store as the UIO for that state.

- Increase length and repeat until all states have a UIO.

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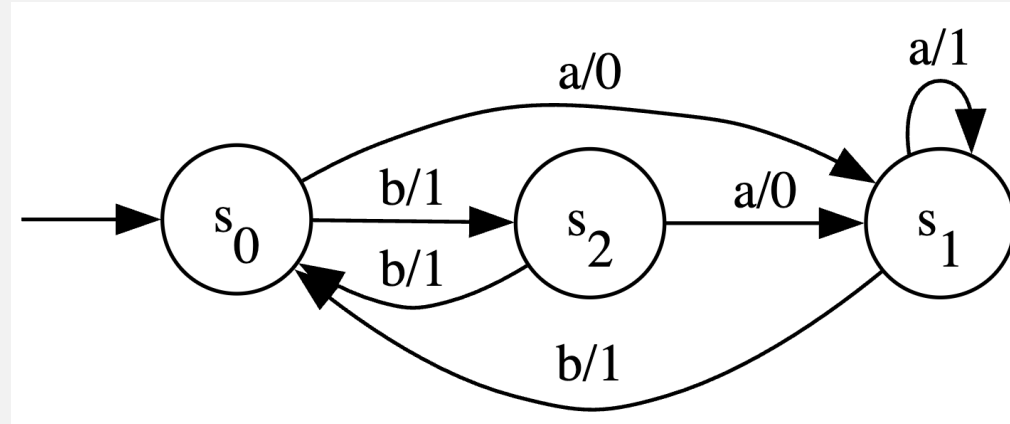
- Increase length and repeat until all states have a UIO.

Easy to understand and implement but ...

- Computationally expensive for large FSMs with large alphabets.

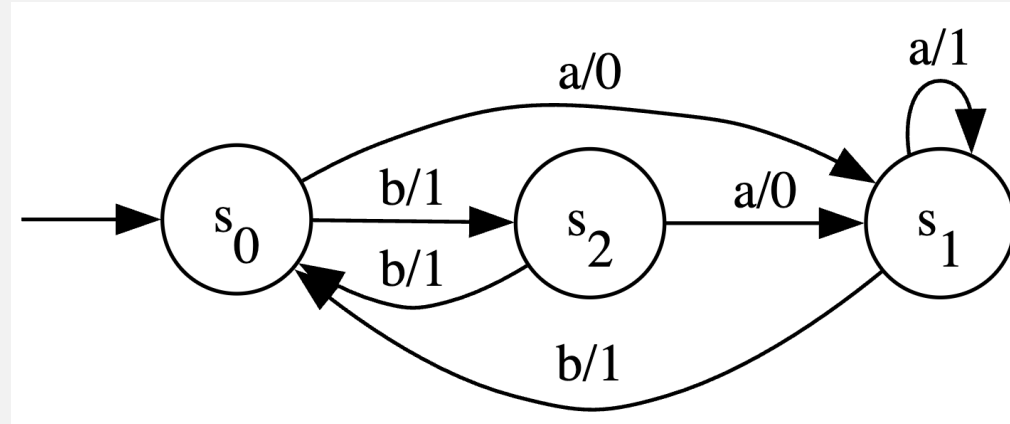
- Finding UIOs is not guaranteed.

Example



Following the brute-force algorithm (building table from left to right):

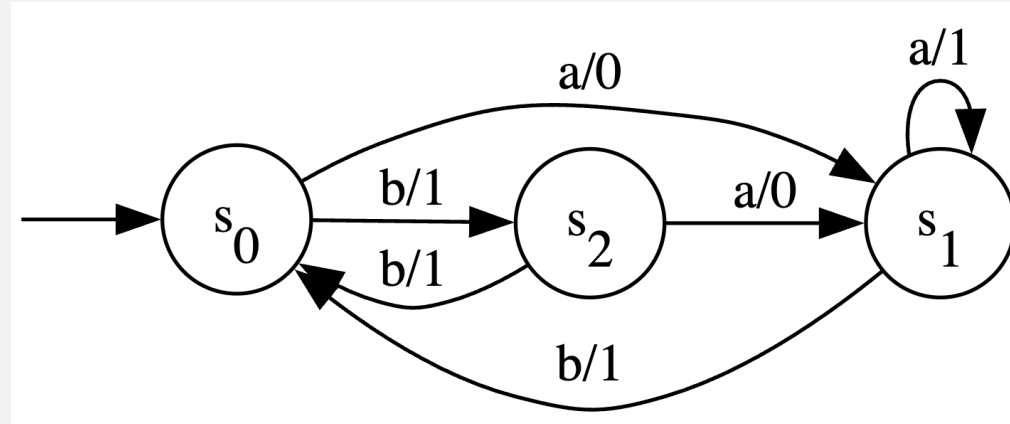
Example



Following the brute-force algorithm (building table from left to right):

State
s0
s1
s2

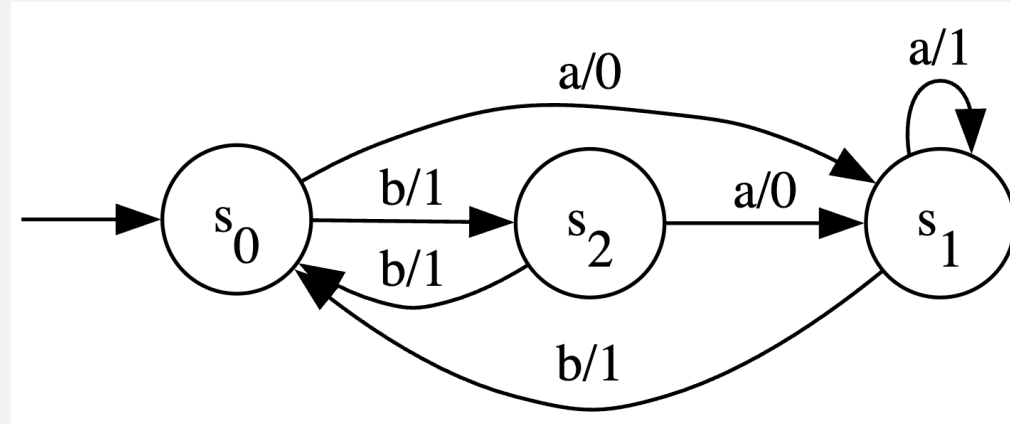
Example



Following the brute-force algorithm (building table from left to right):

State	a
s0	0
s1	1
s2	0

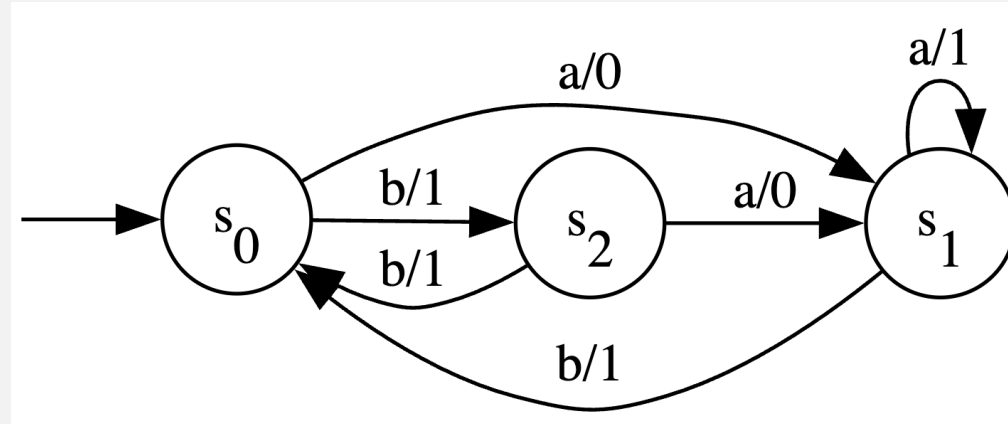
Example



Following the brute-force algorithm (building table from left to right):

State	a	b
s0	0	1
s1	1	1
s2	0	1

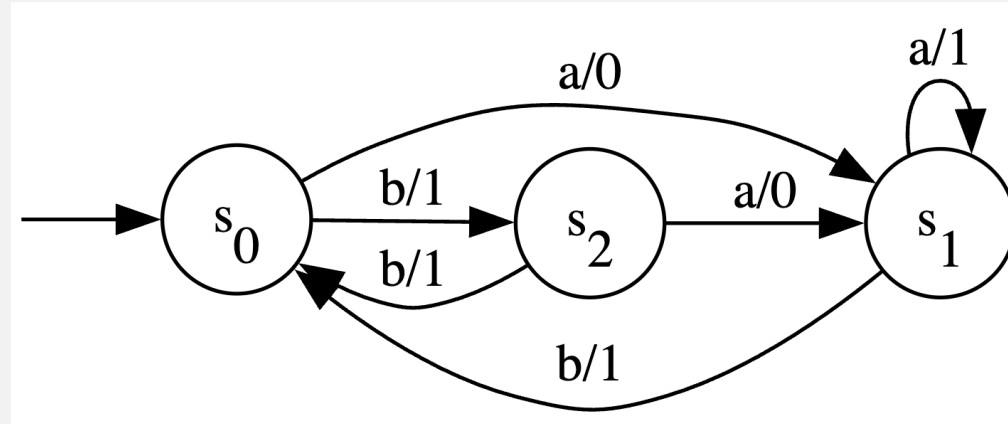
Example



Following the brute-force algorithm (building table from left to right):

State	a	b	aa
s0	0	1	01
s1	1	1	11
s2	0	1	01

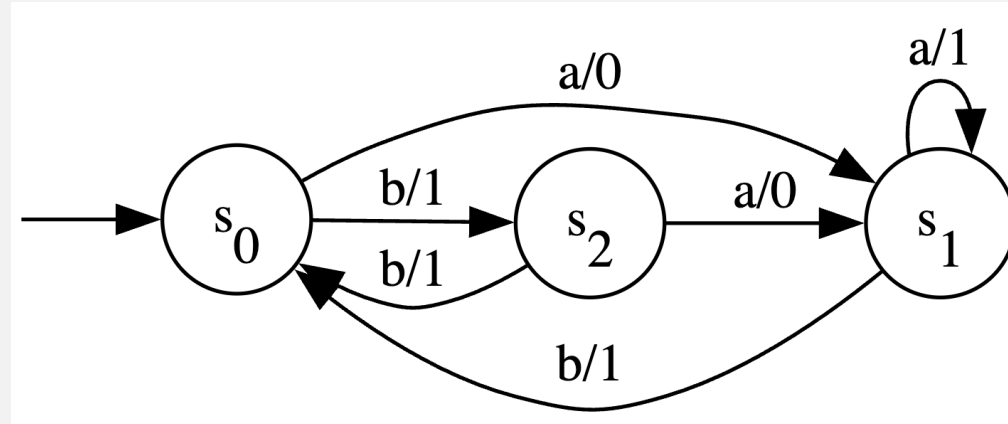
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s1	1	1	11	11
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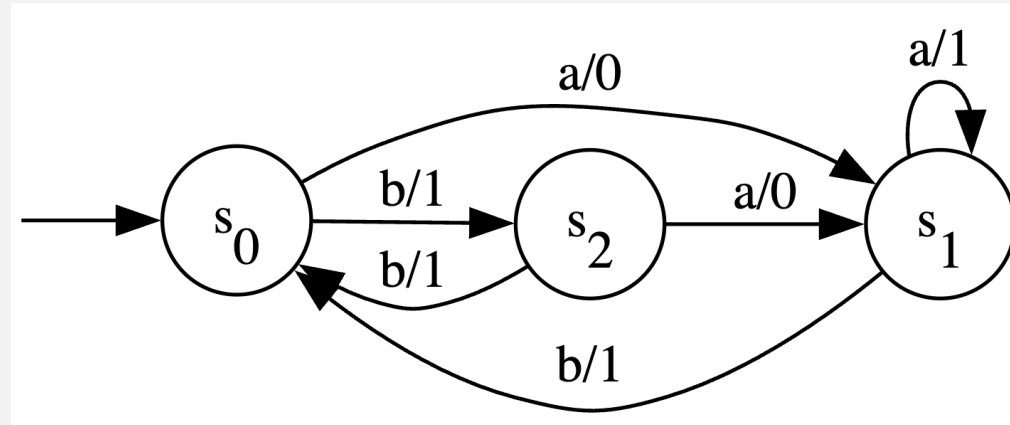
Example



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State	a	b	aa	ab	ba
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s1	1	1	11	11	10
s2	0	1	01	01	10

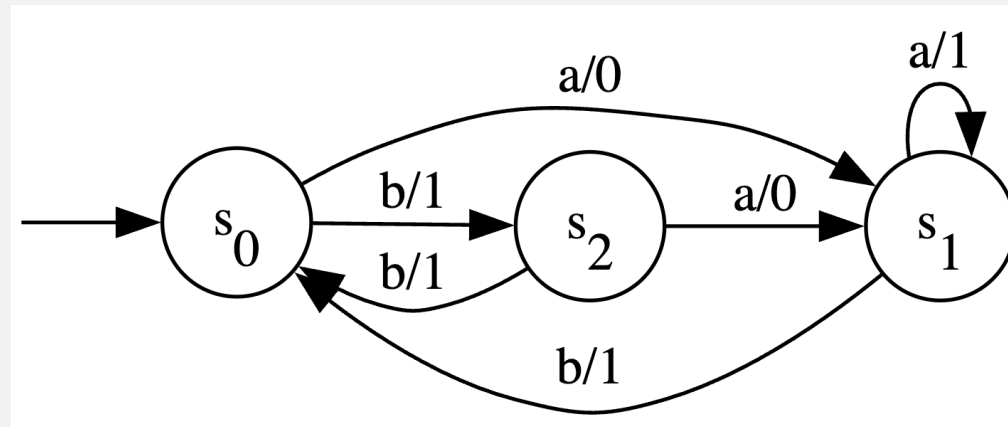
Example



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State	a	b	aa	ab	ba	bb
s0	0	1	01	01	10	11
s1	1	1	11	11	10	11
s2	0	1	01	01	10	11

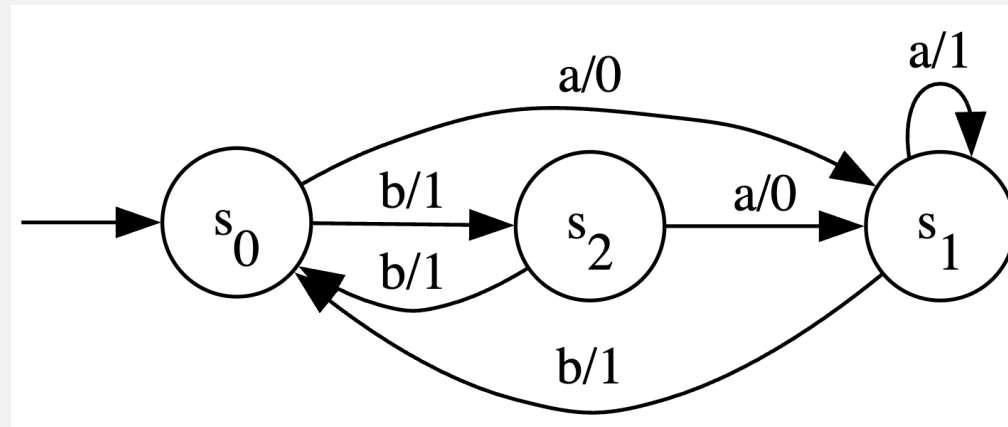
Example



Following the brute-force algorithm (building table from left to right):

State	a	b	aa	ab	ba	bb	aaa
s0	0	1	01	01	10	11	011
s1	1	1	11	11	10	11	111
s2	0	1	01	01	10	11	011

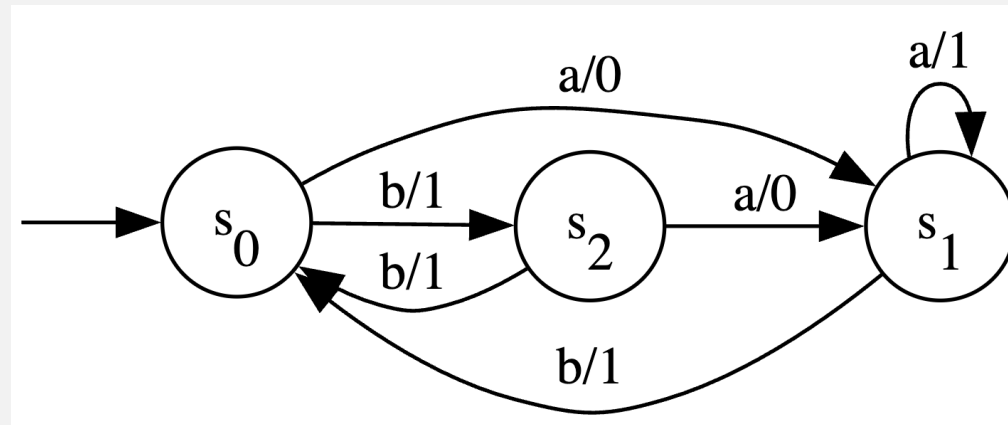
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Example



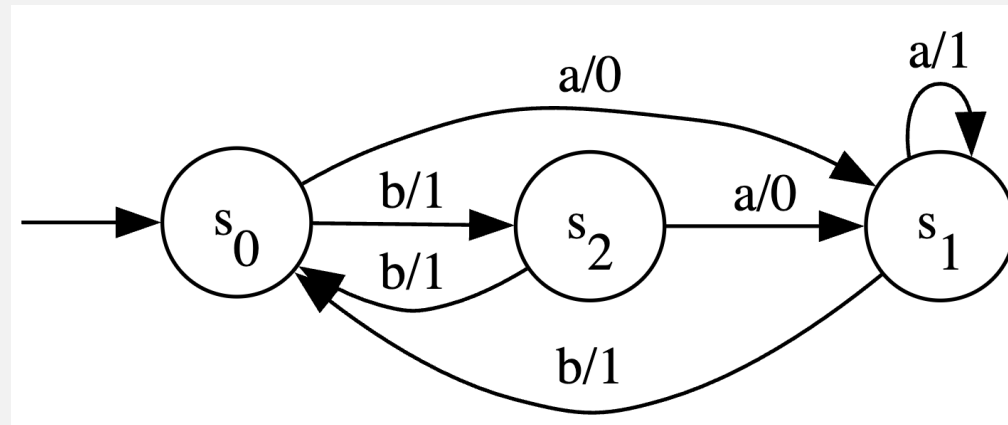
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In the first iteration, we already identify ‘a’ as a UIO for s1.

‘a’ distinguishes s_1 from all other states.

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‘a’ distinguishes s_1 from all other states.

Up to length 2 we cannot find UIOs for states s_0 and s_2 .

Characterising Sets

A **characterising set** W is a set of input sequences that distinguishes each **pair of states**.

For every pair of states s, s' such that $s \neq s'$, we have some $\bar{x} \in W$ such that:

$$\lambda^*(s, \bar{x}) \neq \lambda^*(s', \bar{x})$$

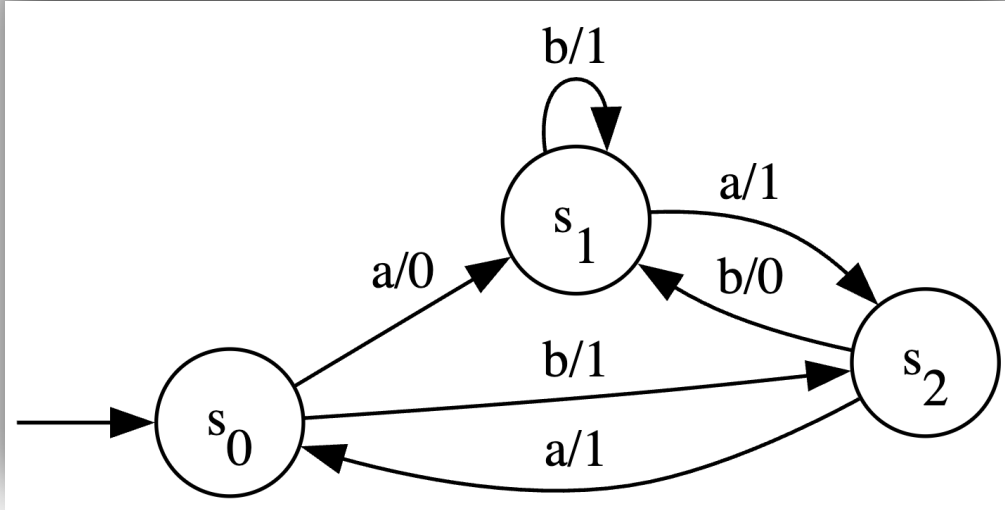
i.e. the input sequence $\bar{x}$ distinguishes these states.

Every minimal FSM with n states has a characterising set W with at most $n - 1$ sequences, of length at most $n - 1$.

There are polynomial time algorithms to generate W sets.

We can minimise every FSM.

Characterising set example



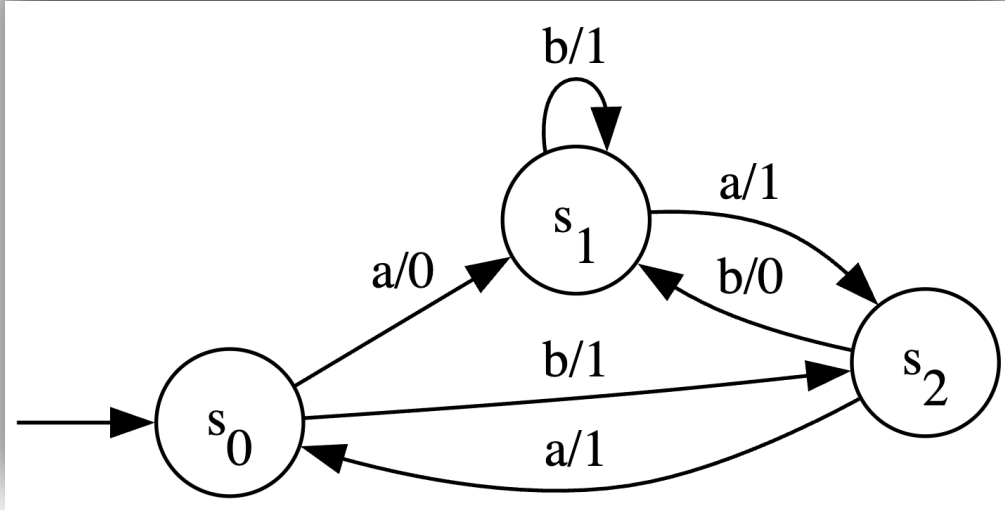
For the above FSM, $W = \{a, b\}$.

If you look at the outputs in response to a and b , they differ for each state.

You can see this in the table.

I.e. For any transition, if we want to check that we are in the correct state, we'll need two tests.

Characterising set example



State
s0
s1
s2

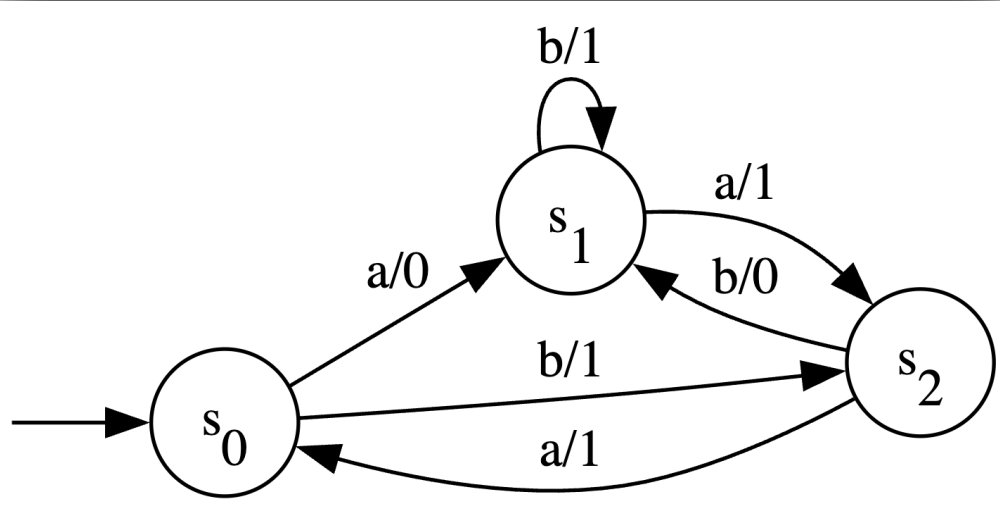
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Characterising set example



State	Response to a
s0	0
s1	1
s2	1

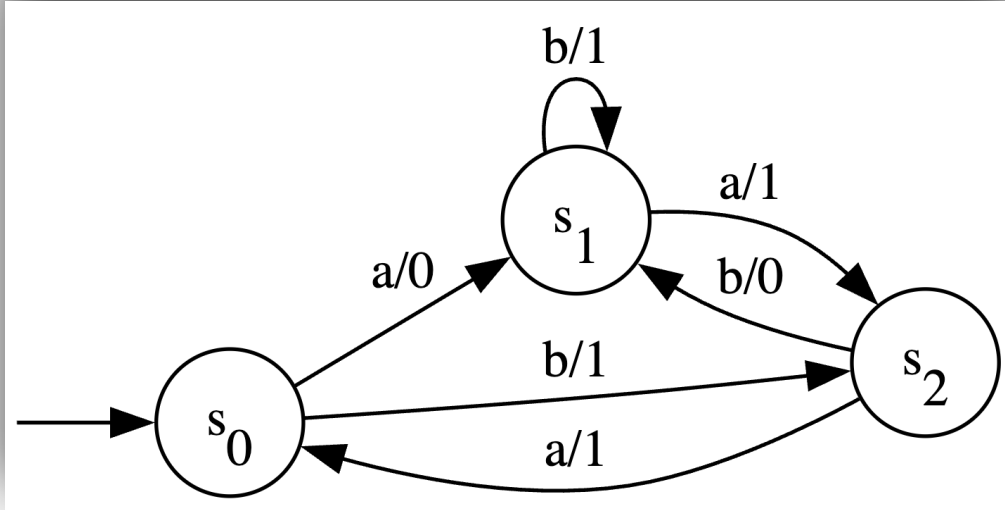
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Characterising set example



State	Response to a	Response to b
s0	0	1
s1	1	1
s2	1	0

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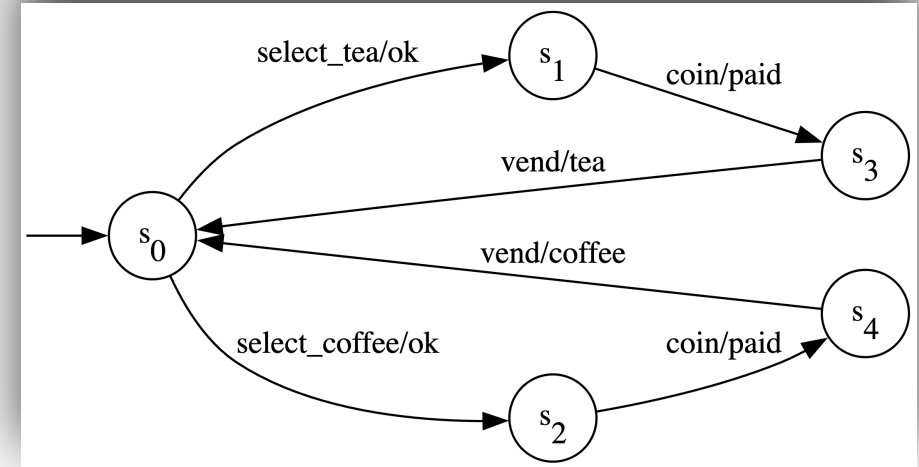
Size of the test set grows exponentially with m .

Testing the Coffee (or Tea) Dispenser with the W-Method

$V = \{ \epsilon, \langle \text{select_tea} \rangle, \langle \text{select_coffee} \rangle, \langle \text{select_tea, coin} \rangle, \langle \text{select_coffee, coin} \rangle \}$

$X = \{ \langle \text{select_tea} \rangle, \langle \text{select_coffee} \rangle, \langle \text{coin} \rangle, \langle \text{vend} \rangle \}$

$W = \{ \langle \text{select_tea, coin, vend} \rangle \}$

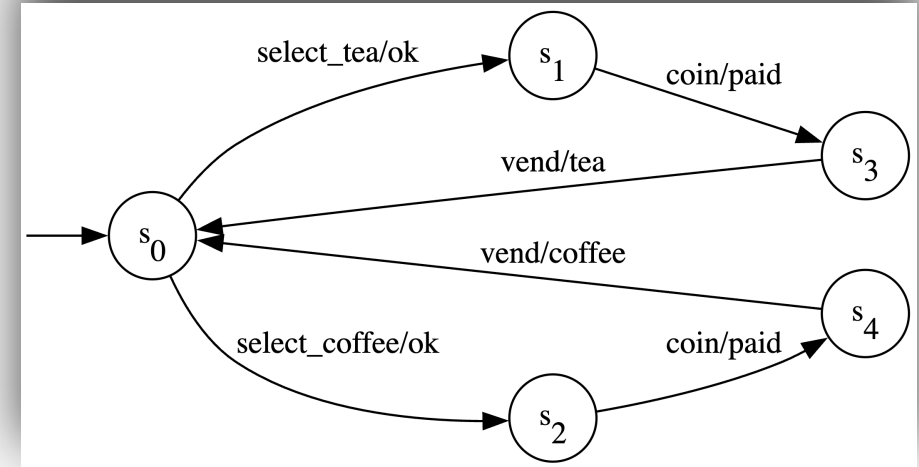


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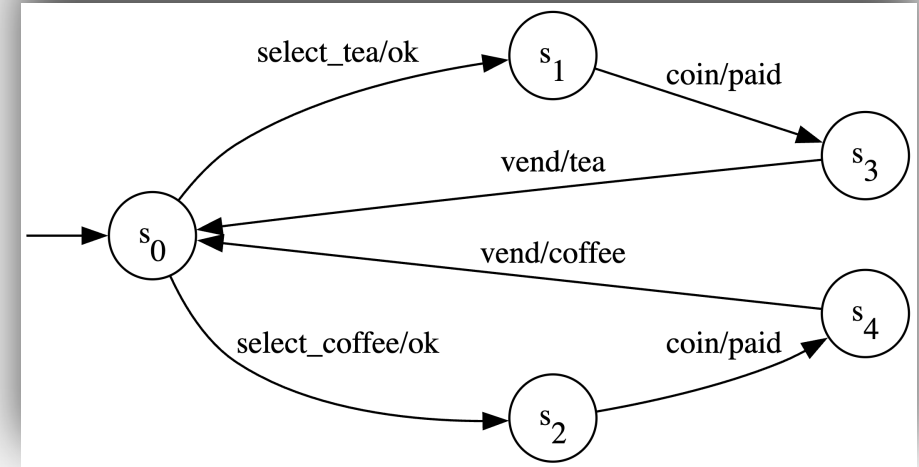
VW
1
2
3
4
5

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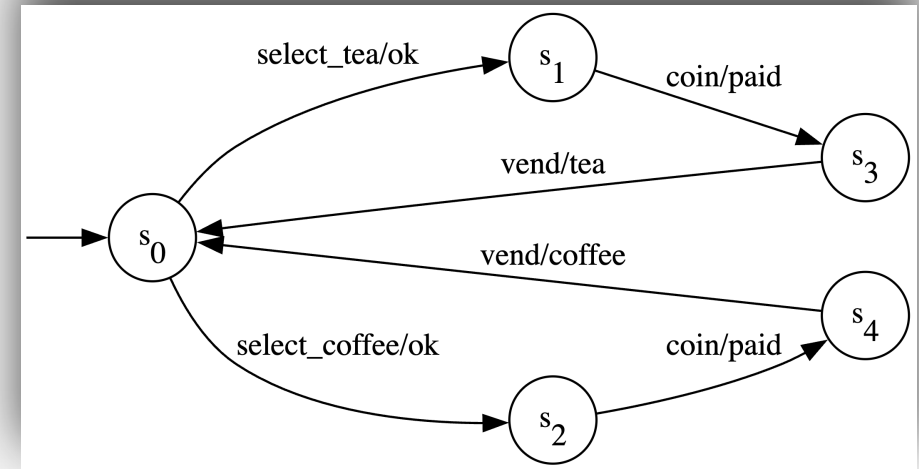
VW	V
1	ϵ
2	$\langle \text{select_tea} \rangle$
3	$\langle \text{select_coffee} \rangle$
4	$\langle \text{select_tea, coin} \rangle$
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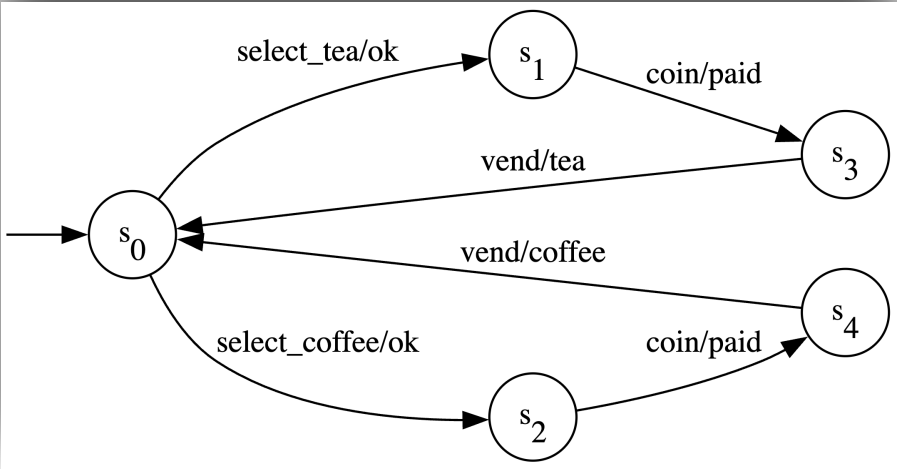
VW	V	W
1	ϵ	$\langle \text{select_tea, coin, vend} \rangle$
2	$\langle \text{select_tea} \rangle$	
3	$\langle \text{select_coffee} \rangle$	
4	$\langle \text{select_tea, coin} \rangle$	
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$W = \{ \langle \text{select_tea, coin, vend} \rangle \}$



VW	V	W	Full sequence
1	ϵ	$\langle \text{select_tea, coin, vend} \rangle$	$\langle \text{select_tea, coin, vend} \rangle$
2	$\langle \text{select_tea} \rangle$		$\langle \text{select_tea, select_tea, coin, vend} \rangle$
3	$\langle \text{select_coffee} \rangle$		$\langle \text{select_coffee, select_tea, coin, vend} \rangle$
4	$\langle \text{select_tea, coin} \rangle$		$\langle \text{select_tea, coin, select_tea, coin, vend} \rangle$
5	$\langle \text{select_coffee, coin} \rangle$		$\langle \text{select_coffee, coin, select_tea, coin, vend} \rangle$

VXW

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

VXW	V
1	ϵ
2	ϵ
3	ϵ
4	ϵ
5	<i><select_tea></i>
6	<i><select_tea></i>
7	<i><select_tea></i>
8	<i><select_tea></i>
9	<i><select_coffee></i>
10	<i><select_coffee></i>
11	<i><select_coffee></i>
12	<i><select_coffee></i>
13	<i><select_tea, coin></i>
14	<i><select_tea, coin></i>
15	<i><select_tea, coin></i>
16	<i><select_tea, coin></i>
17	<i><select_coffee, coin></i>
18	<i><select_coffee, coin></i>
19	<i><select_coffee, coin></i>
20	<i><select_coffee, coin></i>

VXW	V	X
1	ϵ	<i><select_tea></i>
2	ϵ	<i><select_coffee></i>
3	ϵ	<i><coin></i>
4	ϵ	<i><vend></i>
5	<i><select_tea></i>	<i><select_tea></i>
6	<i><select_tea></i>	<i><select_coffee></i>
7	<i><select_tea></i>	<i><coin></i>
8	<i><select_tea></i>	<i><vend></i>
9	<i><select_coffee></i>	<i><select_tea></i>
10	<i><select_coffee></i>	<i><select_coffee></i>
11	<i><select_coffee></i>	<i><coin></i>
12	<i><select_coffee></i>	<i><vend></i>
13	<i><select_tea, coin></i>	<i><select_tea></i>
14	<i><select_tea, coin></i>	<i><select_coffee></i>
15	<i><select_tea, coin></i>	<i><coin></i>
16	<i><select_tea, coin></i>	<i><vend></i>
17	<i><select_coffee, coin></i>	<i><select_tea></i>
18	<i><select_coffee, coin></i>	<i><select_coffee></i>
19	<i><select_coffee, coin></i>	<i><coin></i>
20	<i><select_coffee, coin></i>	<i><vend></i>

VXW	V	X	W
1	ϵ	<i><select_tea></i>	<i><select_tea, coin, vend></i>
2	ϵ	<i><select_coffee></i>	
3	ϵ	<i><coin></i>	
4	ϵ	<i><vend></i>	
5	<i><select_tea></i>	<i><select_tea></i>	
6	<i><select_tea></i>	<i><select_coffee></i>	
7	<i><select_tea></i>	<i><coin></i>	
8	<i><select_tea></i>	<i><vend></i>	
9	<i><select_coffee></i>	<i><select_tea></i>	
10	<i><select_coffee></i>	<i><select_coffee></i>	
11	<i><select_coffee></i>	<i><coin></i>	
12	<i><select_coffee></i>	<i><vend></i>	
13	<i><select_tea, coin></i>	<i><select_tea></i>	
14	<i><select_tea, coin></i>	<i><select_coffee></i>	
15	<i><select_tea, coin></i>	<i><coin></i>	
16	<i><select_tea, coin></i>	<i><vend></i>	
17	<i><select_coffee, coin></i>	<i><select_tea></i>	
18	<i><select_coffee, coin></i>	<i><select_coffee></i>	
19	<i><select_coffee, coin></i>	<i><coin></i>	
20	<i><select_coffee, coin></i>	<i><vend></i>	

VXW	V	X	W	Full sequence
1	ε	<i><select_tea></i>	<i><select_tea, coin, vend></i>	<i><select_tea, coin, vend></i>
2	ε	<i><select_coffee></i>		<i><select_coffee, select_tea, coin, vend></i>
3	ε	<i><coin></i>		<i><coin, select_tea, coin, vend></i>
4	ε	<i><vend></i>		<i><vend, select_tea, coin, vend></i>
5	<i><select_tea></i>	<i><select_tea></i>		<i><select_tea, select_tea, select_tea, coin, vend></i>
6	<i><select_tea></i>	<i><select_coffee></i>		<i><select_tea, select_coffee, select_tea, coin, vend></i>
7	<i><select_tea></i>	<i><coin></i>		<i><select_tea, coin, select_tea, coin, vend></i>
8	<i><select_tea></i>	<i><vend></i>		<i><select_tea, vend, select_tea, coin, vend></i>
9	<i><select_coffee></i>	<i><select_tea></i>		<i><select_coffee, select_tea, select_tea, coin, vend></i>
10	<i><select_coffee></i>	<i><select_coffee></i>		<i><select_coffee, select_coffee, select_tea, coin, vend></i>
11	<i><select_coffee></i>	<i><coin></i>		<i><select_coffee, coin, select_tea, coin, vend></i>
12	<i><select_coffee></i>	<i><vend></i>		<i><select_coffee, vend, select_tea, coin, vend></i>
13	<i><select_tea, coin></i>	<i><select_tea></i>		<i><select_tea, coin, select_tea, select_tea, coin, vend></i>
14	<i><select_tea, coin></i>	<i><select_coffee></i>		<i><select_tea, coin, select_coffee, select_tea, coin, vend></i>
15	<i><select_tea, coin></i>	<i><coin></i>		<i><select_tea, coin, coin, select_tea, coin, vend></i>
16	<i><select_tea, coin></i>	<i><vend></i>		<i><select_tea, coin, vend, select_tea, coin, vend></i>
17	<i><select_coffee, coin></i>	<i><select_tea></i>		<i><select_coffee, coin, select_tea, select_tea, coin, vend></i>
18	<i><select_coffee, coin></i>	<i><select_coffee></i>		<i><select_coffee, coin, select_coffee, select_tea, coin, vend></i>
19	<i><select_coffee, coin></i>	<i><coin></i>		<i><select_coffee, coin, coin, select_tea, coin, vend></i>
20	<i><select_coffee, coin></i>	<i><vend></i>		<i><select_coffee, coin, vend, select_tea, coin, vend></i>

VXW	V	X	W	Full sequence
1	ϵ	<i><select_tea></i>		<i><select_tea,coin,vend></i>
2	ϵ	<i><select_coffee></i>		<i><select_coffee, select_tea, coin, vend></i>
3	ϵ	<i><coin></i>		<i><coin, select_tea, coin, vend></i>
4	ϵ	<i><vend></i>		<i><vend, select_tea, coin, vend></i>
5	<i><select_tea></i>	<i><select_tea></i>		<i><select_tea, select_tea, select_tea, coin, vend></i>
6	<i><select_tea></i>	<i><select_coffee></i>		<i><select_tea, select_coffee, select_tea, coin, vend></i>
7	<i><select_tea></i>	<i><coin></i>		<i><select_tea, coin, select_tea, coin, vend></i>
8	<i><select_tea></i>		VW contains 5 tests VXW contains 20 tests Full test set contains all 25 tests	<i><vend, select_tea, coin, vend></i>
9	<i><select_coffee></i>	<i><select_tea></i>		<i><select_coffee, select_tea, coin, vend></i>
10	<i><select_coffee></i>	<i><select_coffee></i>		<i><select_coffee, select_tea, coin, vend></i>
11	<i><select_coffee></i>			<i><coin, select_tea, coin, vend></i>
12	<i><select_coffee></i>			<i><vend, select_tea, coin, vend></i>
13	<i><select_tea, coin></i>	<i><select_tea></i>		<i><select_tea, coin, select_tea, select_tea, coin, vend></i>
14	<i><select_tea, coin></i>	<i><select_coffee></i>		<i><select_tea, coin, select_coffee, select_tea, coin, vend></i>
15	<i><select_tea, coin></i>	<i><coin></i>		<i><select_tea, coin, coin, select_tea, coin, vend></i>
16	<i><select_tea, coin></i>	<i><vend></i>		<i><select_tea, coin, vend, select_tea, coin, vend></i>
17	<i><select_coffee, coin></i>	<i><select_tea></i>		<i><select_coffee, coin, select_tea, select_tea, coin, vend></i>
18	<i><select_coffee, coin></i>	<i><select_coffee></i>		<i><select_coffee, coin, select_coffee, select_tea, coin, vend></i>
19	<i><select_coffee, coin></i>	<i><coin></i>		<i><select_coffee, coin, coin, select_tea, coin, vend></i>
20	<i><select_coffee, coin></i>	<i><vend></i>		<i><select_coffee, coin, vend, select_tea, coin, vend></i>

A hand holding a pen over a document with a blue overlay.

Wrapping up

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Where do the models come from?

Particularly powerful in Model-Driven Development, where models exist anyway.

Several tools that can reverse-engineer models from input/output observations.